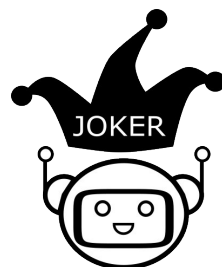


Pun Intended: Multi-Agent Translation of Wordplay with Contrastive Learning and Phonetic-Semantic Embeddings

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Best of Labs 2025 · CLEF 2026, Jena



Can AI Translate a Joke?

*“In ancient times, seagoing vessels were much more fuel efficient.
They got thousands of miles to the **galleon**.”*

Baseline

« Autrefois, avec une **pâte** infaillible, les boulangers n'étaient jamais dans le **pâté**. »

Has French wordplay, but no longer translates or recreates the maritime source.

Guided

« Autrefois, les marins faisaient des milliers de lieues au **gallon**. »

Translates literally, eliminates the wordplay.

Evaluator

« Autrefois, les navires faisaient des milliers de milles par **gallon**... enfin, par **galion** ! »

Preserves both the semantic context and humor of the source.

Can we preserve both semantic content and the wordplay it supports?

One Task

THE CHALLENGE

- Pun translation must preserve both meaning and wordplay
- Puns rely on semantic ambiguity and incongruity; LLMs are trained on regular patterns
- An equivalent pun rarely exists in the target language: new wordplay must be created
- Humor translation is among the hardest tasks, even for professionals

THE OPPORTUNITY

- LLMs bring broad linguistic and cultural knowledge from web-scale corpora
- Computational approaches that were previously impractical are now feasible
- Recent pun-generation work models ambiguity, phonetic similarity and iterative refinement

Two Contributions

CONTRIBUTION 1

Prioritizing functional equivalence over literal correspondence produces stronger human evaluation for cross-lingual wordplay translation.

CONTRIBUTION 2 · THREE PROGRESSIVELY MORE SOPHISTICATED LLM-BASED APPROACHES

**Discriminator-guided
generation**

Baseline



**Explicit phonetic-semantic
guidance**

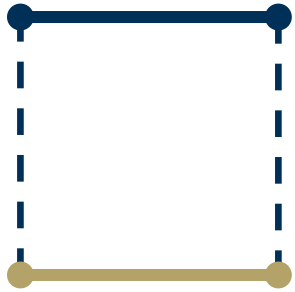
Guided chain-of-thought



**Iterative multi-agent
evaluation**

Multi-agent

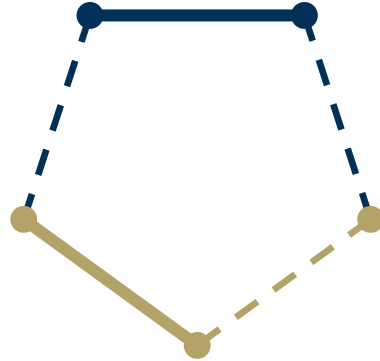
Close the circle between two meanings (Low 2011)



Square

0 leaps

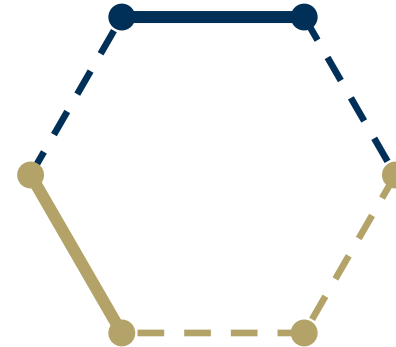
in verse / inverse



Pentagon

1 leap

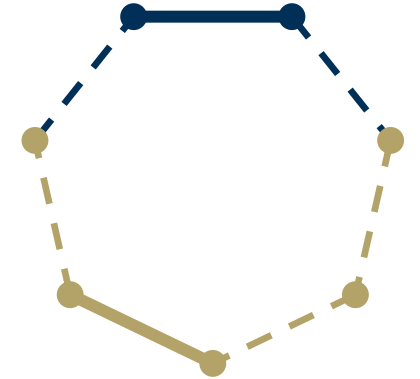
Bugger off for bugger all



Hexagon

2 leaps

Life depends on the liver



Circle

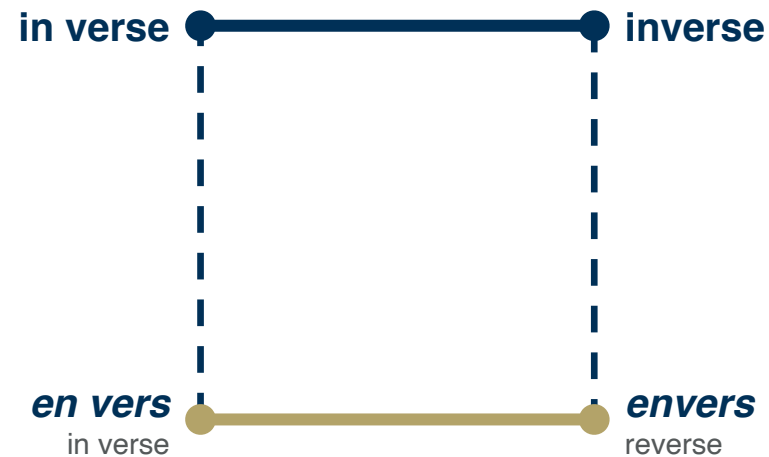
3 leaps

purpose / porpoise

Translate both meanings, then leap through related words until two French words make a pun.

— source pun - - translation - - semantic leap — target pun

The square: the pun survives direct translation



SOURCE

“If he writes in verse, take the inverse meaning.”

SEARCH PATH

- 1 in verse → *en vers*
- 2 inverse → *envers*
- 3 *en vers* ≈ *envers*: the pun is already there

RESULT

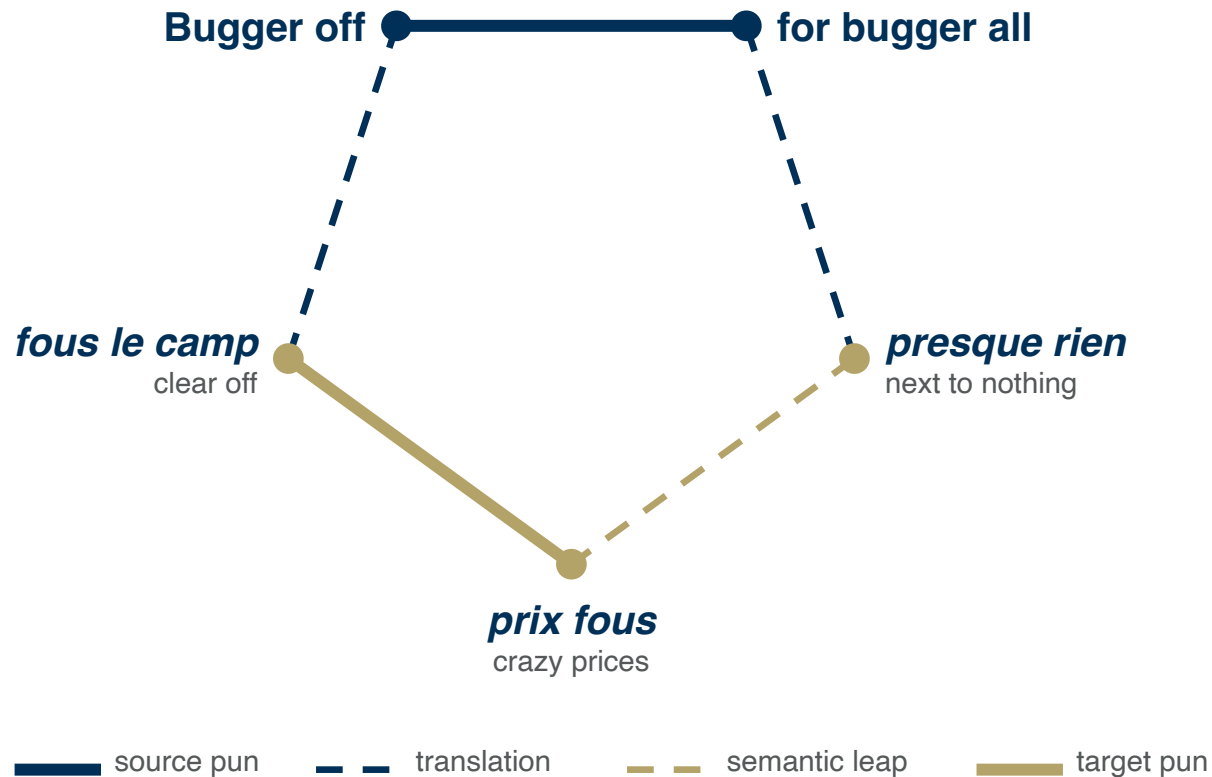
« *S’il écrit en vers, le sens sera l’envers.* »

Shared puns like this are rare, even between related languages.

— source pun - - translation - - semantic leap — target pun

Adapted from Low, P. A. (2011). *Translating Jokes and Puns. Perspectives* 19(1), Figs. 1–4.

The pentagon: one semantic leap



SOURCE

Air New Zealand: “Bugger off for bugger all.”

SEARCH PATH

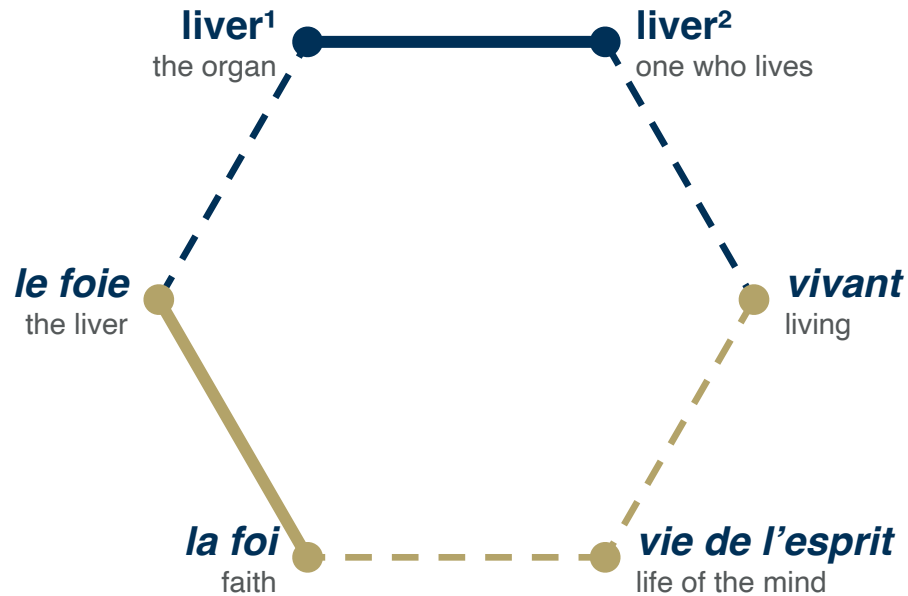
- 1 Bugger off → *Foutez le camp !*
- 2 for bugger all → *pour presque rien* (no pun yet)
- 3 Leap: next to nothing → *les prix sont fous*
- 4 *fous* (crazy) = *fous* (clear off): circle closed

RESULT

« *Nos prix sont fous, fous... fous le camp !* »

Low also suggests a square: « *Foutre le camp pour foutre rien* ».

The hexagon: two semantic leaps



— source pun - - translation - - semantic leap — target pun

SOURCE

Medical epigram: “Life depends on the liver.”

SEARCH PATH

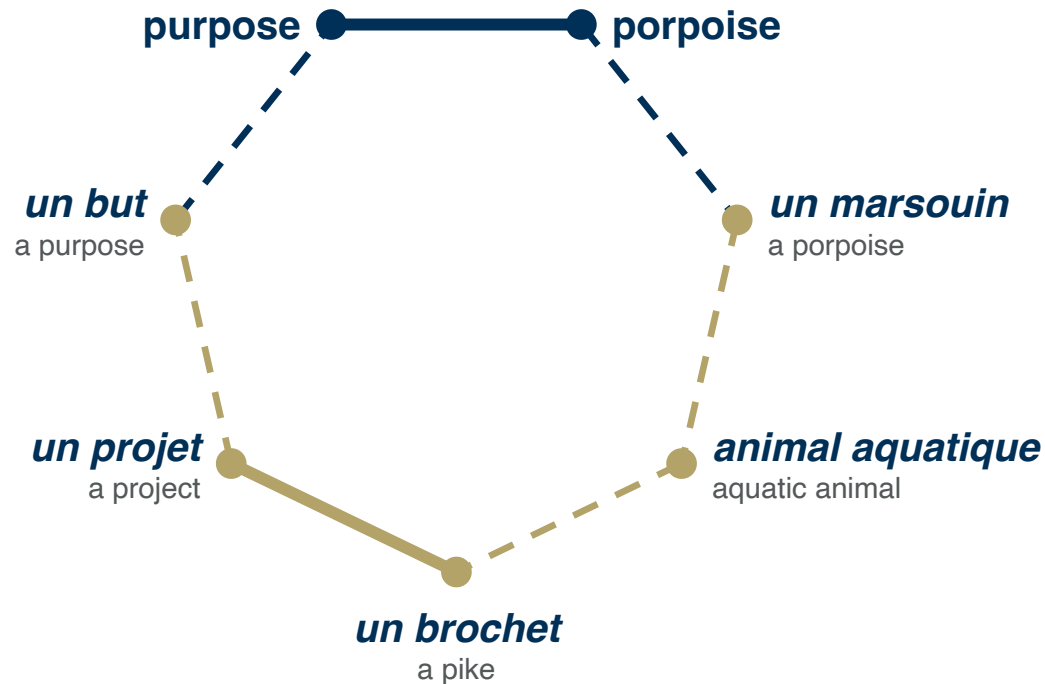
- 1 liver (organ) → *le foie*
- 2 liver (one who lives) → *vivant*
- 3 Leap: living → *la vie de l'esprit* (life of the mind)
- 4 Leap: life of the mind → *la foi* (faith)
- 5 *foi* sounds like *foie*: circle closed

RESULT

« *La vie est une question de foie.* »

Life is a question of liver / of faith.

The circle: three or more leaps (Alice in Wonderland)



— source pun - - translation - - semantic leap — target pun

SOURCE

Mock Turtle: “With what porpoise?” Alice: “Don’t you mean ‘purpose’?”

SEARCH PATH

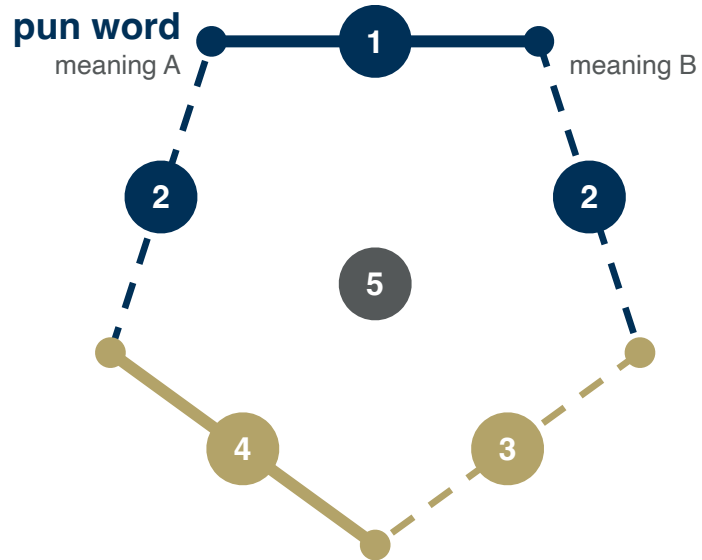
- 1 purpose → *un but* → synonym *un projet*
- 2 porpoise → *un marsouin* → *animal aquatique*
- 3 Leap within the fish pool: *un brochet* (pike)
- 4 *brochet* ≈ *projet*: circle closed

RESULT

« *Avec quel brochet ?* »

Alice: « projet », et non « brochet » ? (Henri Bué, first French Alice)

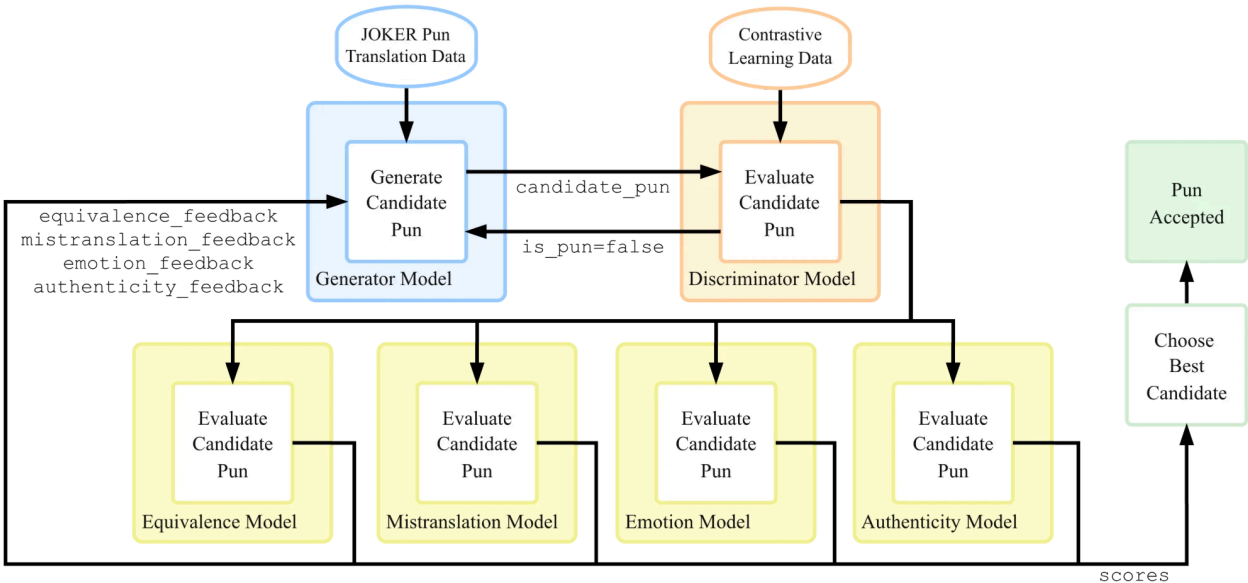
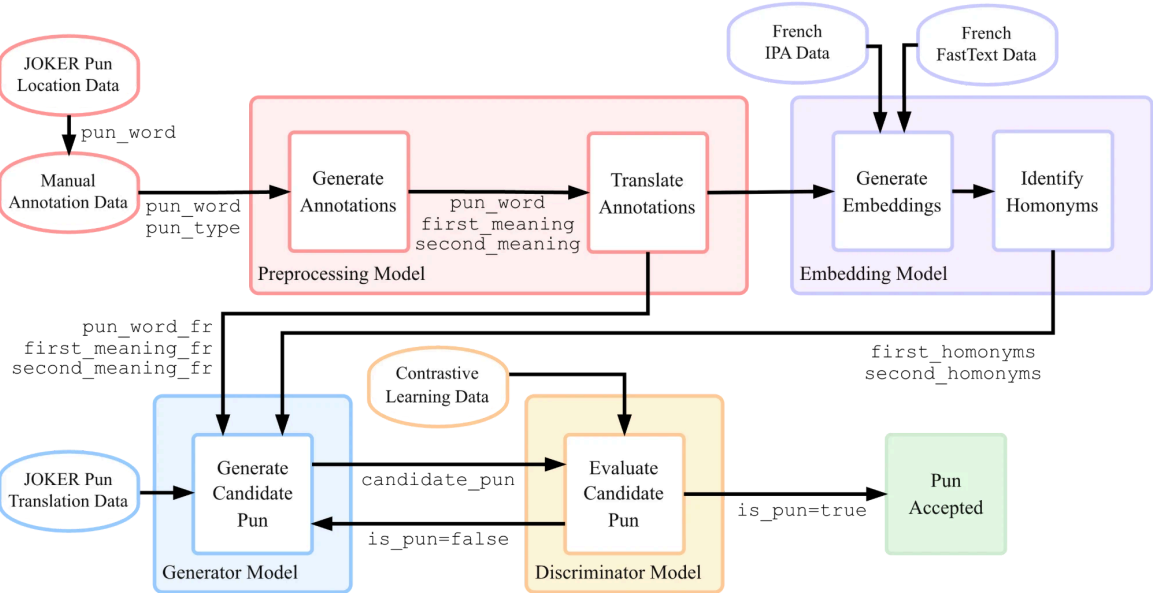
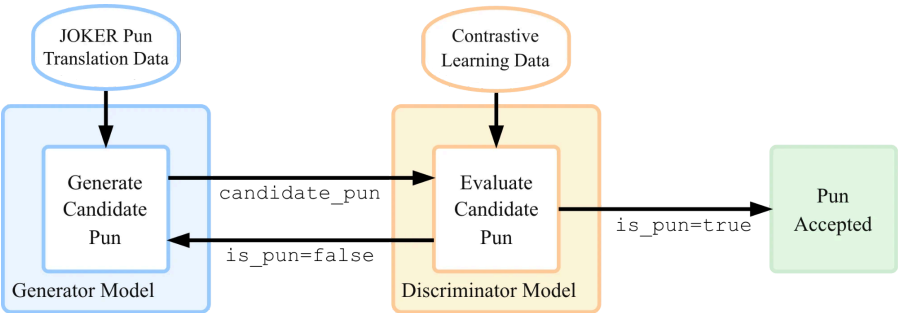
From Low's polygon to a retrieval pipeline



- 1 English pun edge**
LLM pun identification: pun word, homographic / homophonic, both meanings (Gemini 2.5 Pro)
- 2 Translation edges**
LLM translation of the pun word and a synonym list for each meaning (o4-mini)
- 3 Semantic leap**
French FastText semantic embeddings (300-d)
- 4 French pun edge**
French phonetic embeddings: IPA → PanPhon features → BiLSTM (300-d)
- 5 “Is it a joke?”**
Contrastive discriminator (Gemini 2.5 Flash, 25 + 25 few-shot)

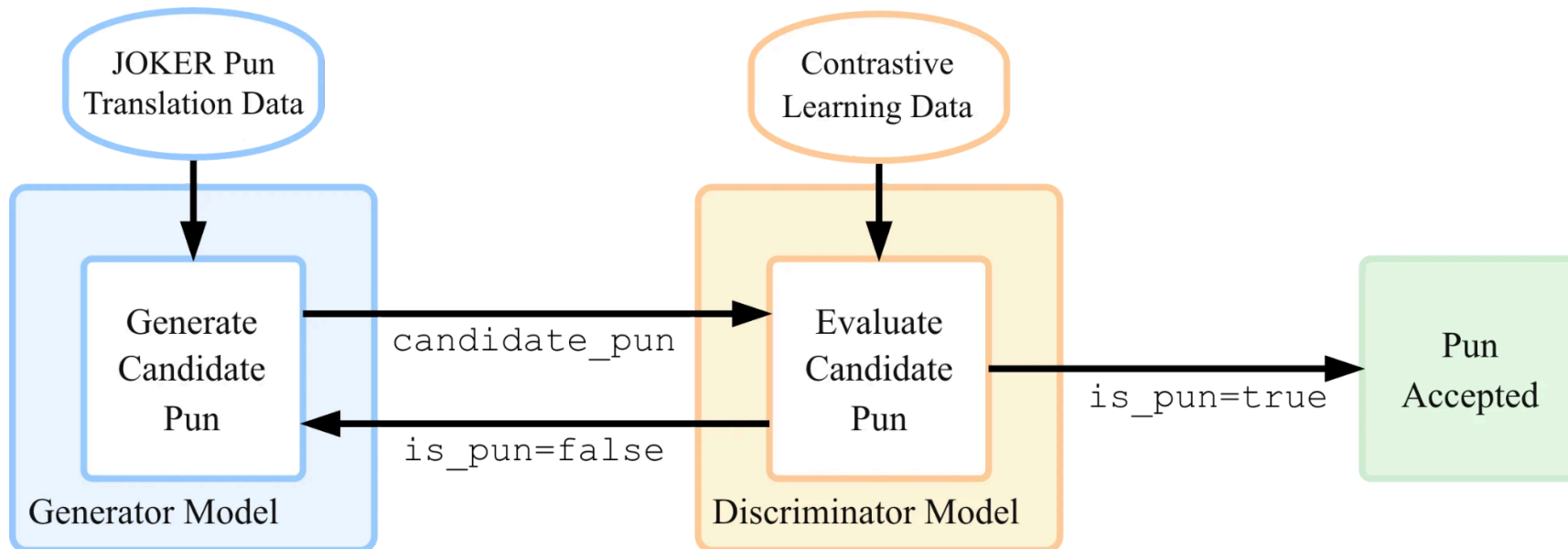
— source pun - - translation - - semantic leap — target pun

Three Approaches



Approach 1: Baseline

Generate first. Reject non-puns.

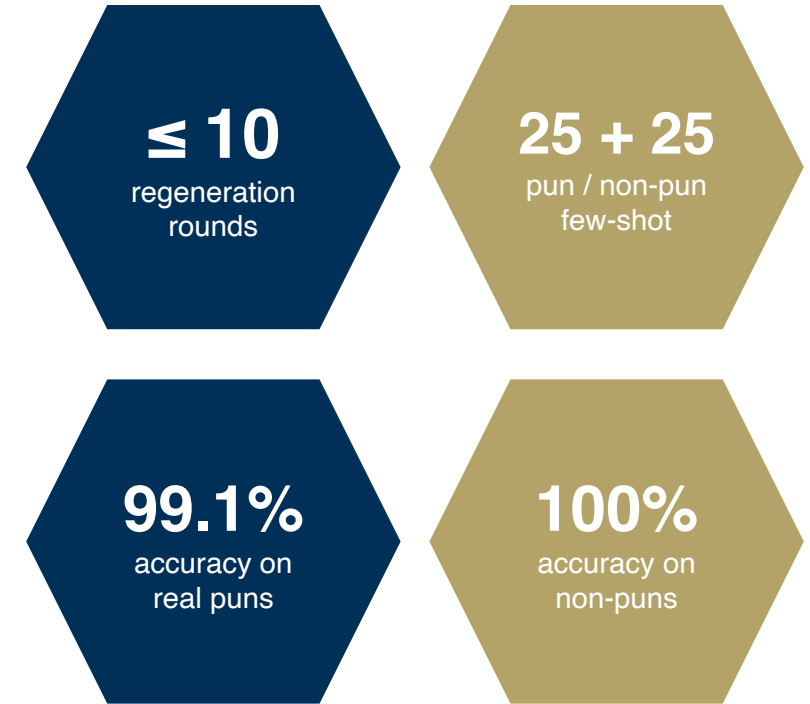


25 + 25 contrastive examples · regenerate $\leq 10\times$ · discriminator (n = 450): 99.1% positive, 100% negative

Baseline: Generator and contrastive discriminator

Prompt to **create** a French pun, not to translate: emphasize native fluency and humor, use a homonym whose two senses both fit the context (after Mittal et al., 2022).

Generators: Mistral Medium 2505, o4-mini, Discriminator: Gemini 2.5 Flash



Discriminator tested on 450 hand-labeled examples

Contrastive discrimination example

POSITIVE · FRENCH PUN

« **Ça fait douze au carré** », dit-il carrément.

carré / carrément: the wordplay carries the joke

NEGATIVE · GENERATED NON-PUN TWIN

« **Ça fait cent quarante-quatre** », dit-il franchement.

same content, same length, no wordplay

1

For every French reference pun, Gemini 2.5 Flash writes a non-pun twin

2

An independent prompt checks the twin; any leftover wordplay → regenerate

3

Sample 25 puns + 25 twins as few-shot examples for the discriminator

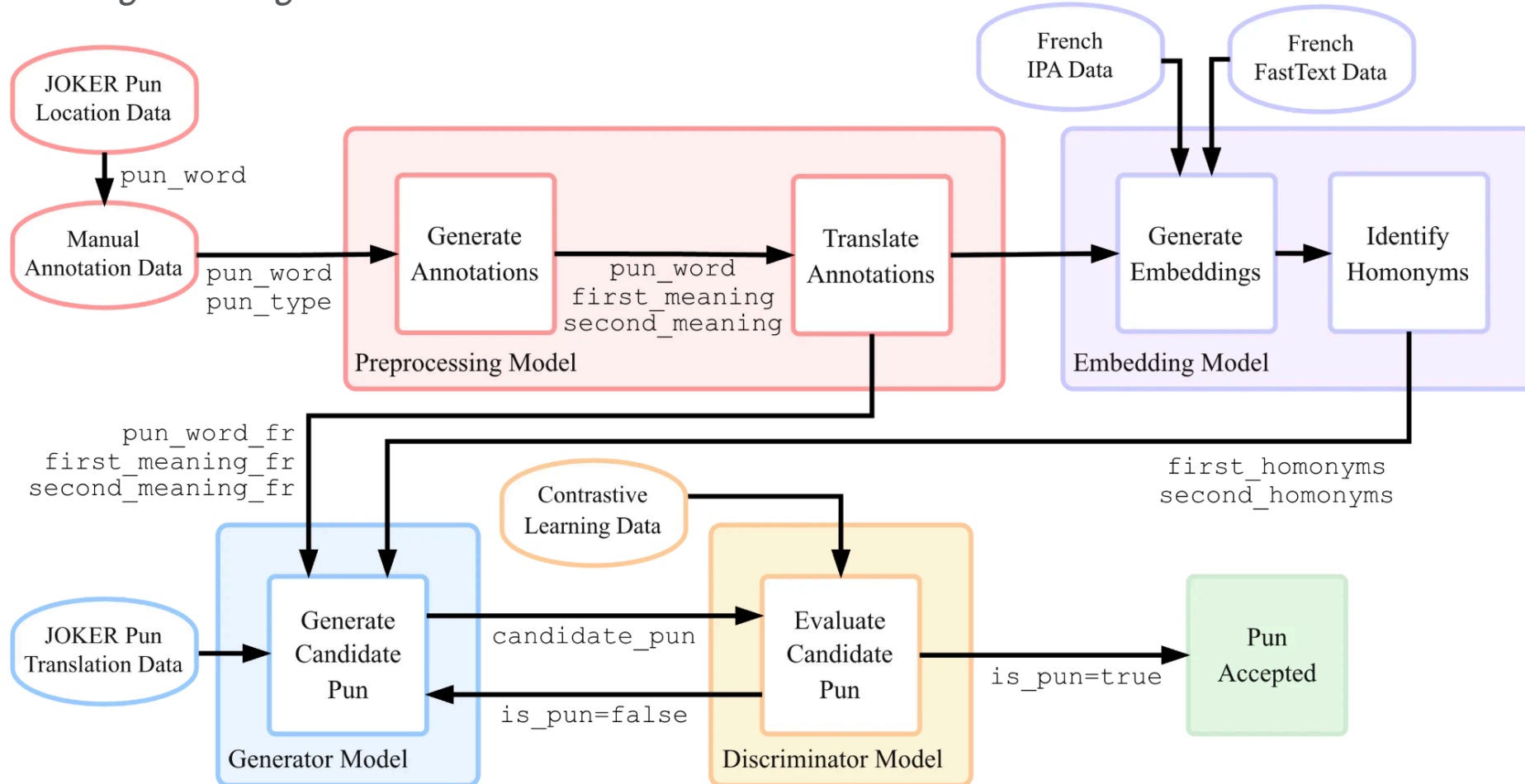
4

Candidate judged “not a pun”
→ back to the generator

Positive: our system's output for en_160. Negative: illustrative example of the generated twin.

Approach 2: Guided chain-of-thought

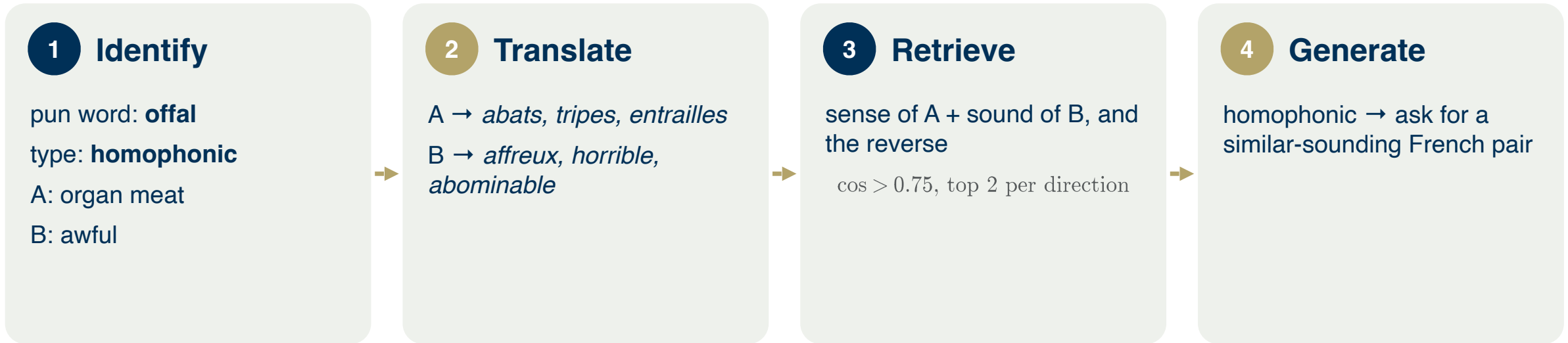
Decompose before generating



Pun localization: F1 .97 · Pun type: F1 .89

Walkthrough: one pun through the guided pipeline

en_40 “The class took a field trip to a meat processing plant, but what they saw was just offal.”



OUTPUT « *Un vrai trip de tripes !* » tripes (guts) ≈ trip (slang: an experience)

Output is the guided system's submitted translation; intermediate values illustrate each stage.

Phonetic-semantic retrieval

Where can the two meanings collide in French?

PHONETIC SPACE · 300-d BiLSTM

SEMANTIC SPACE · 300-d FastText

Lexique → IPA → PanPhon features → phoneme bigrams

pre-trained French FastText

$$J_{\text{phon}}(a, b) = \frac{|F(a) \cap F(b)|}{|F(a) \cup F(b)|}$$

where a, b are phoneme bigrams

Following Sharma et al.; adapted to French

Querying in both directions



$$\cos(\mathbf{w}_{\text{sem}}, \mathbf{S}) > 0.75 \quad \text{and} \quad \cos(\mathbf{w}_{\text{phon}}, \mathbf{P}) > 0.75$$

keep the top 2 candidates per direction

Guided generation: three strategies by pun type

1

Homographic, French word still ambiguous

Build the translation around the translated pun word.

wrapped / rapt → **emballée**
emballer = to wrap / to thrill (Low, 2011)

2

Homographic, no French equivalent

Pick a French homonym whose two senses best approximate both meanings.

Retrieved candidates rank the homonyms

3

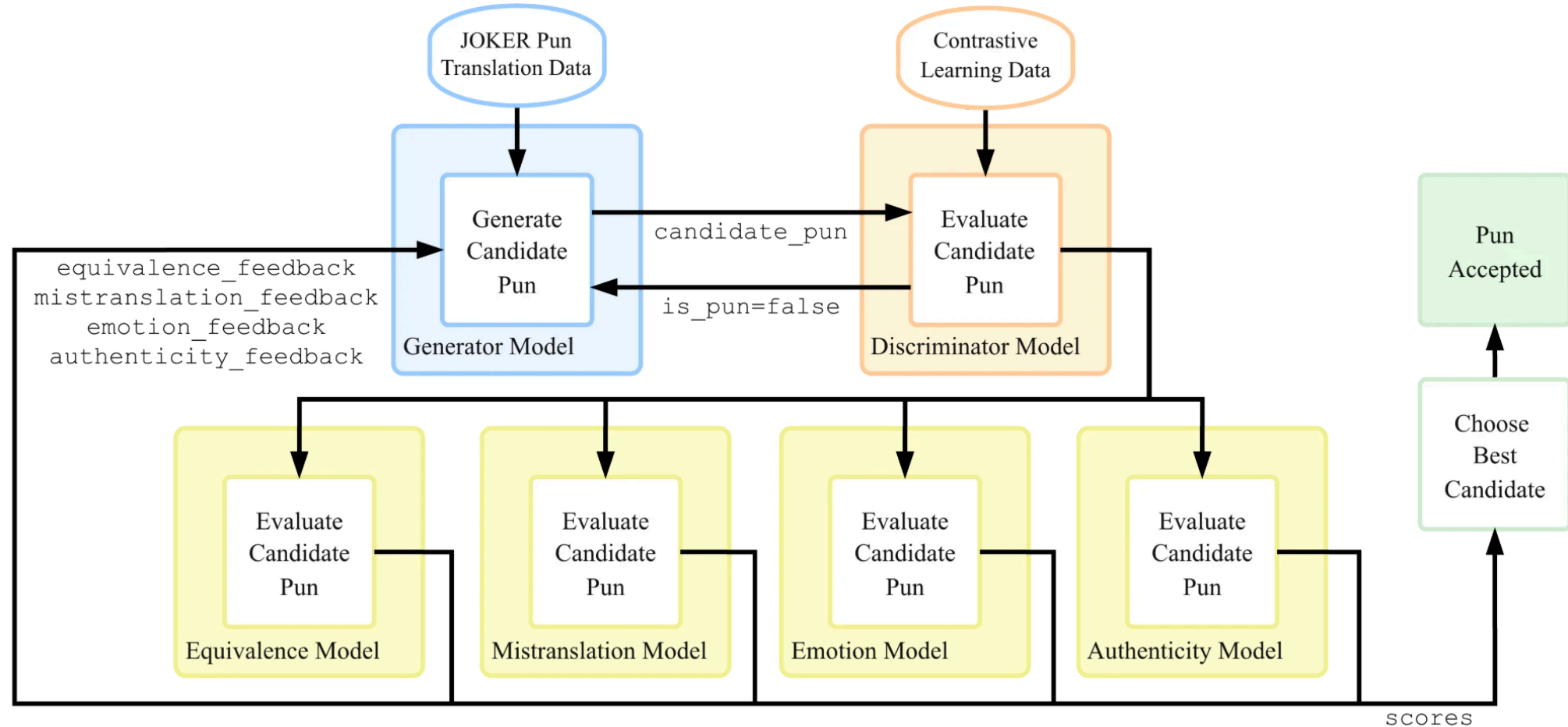
Homophonic

Generate two similar-sounding French words, one per meaning.

offal / awful → **tripes** / **trip**
guided system, en_40

Approach 3: Multi-agent evaluation and refinement

Generate freely. Evaluate structurally. Revise.



Equivalence 0–2 · Quality 0–2 · Emotion 0–1 · Authenticity 0–4
stop at mean ≥ 2.0 or 5 iterations

Four Evaluators

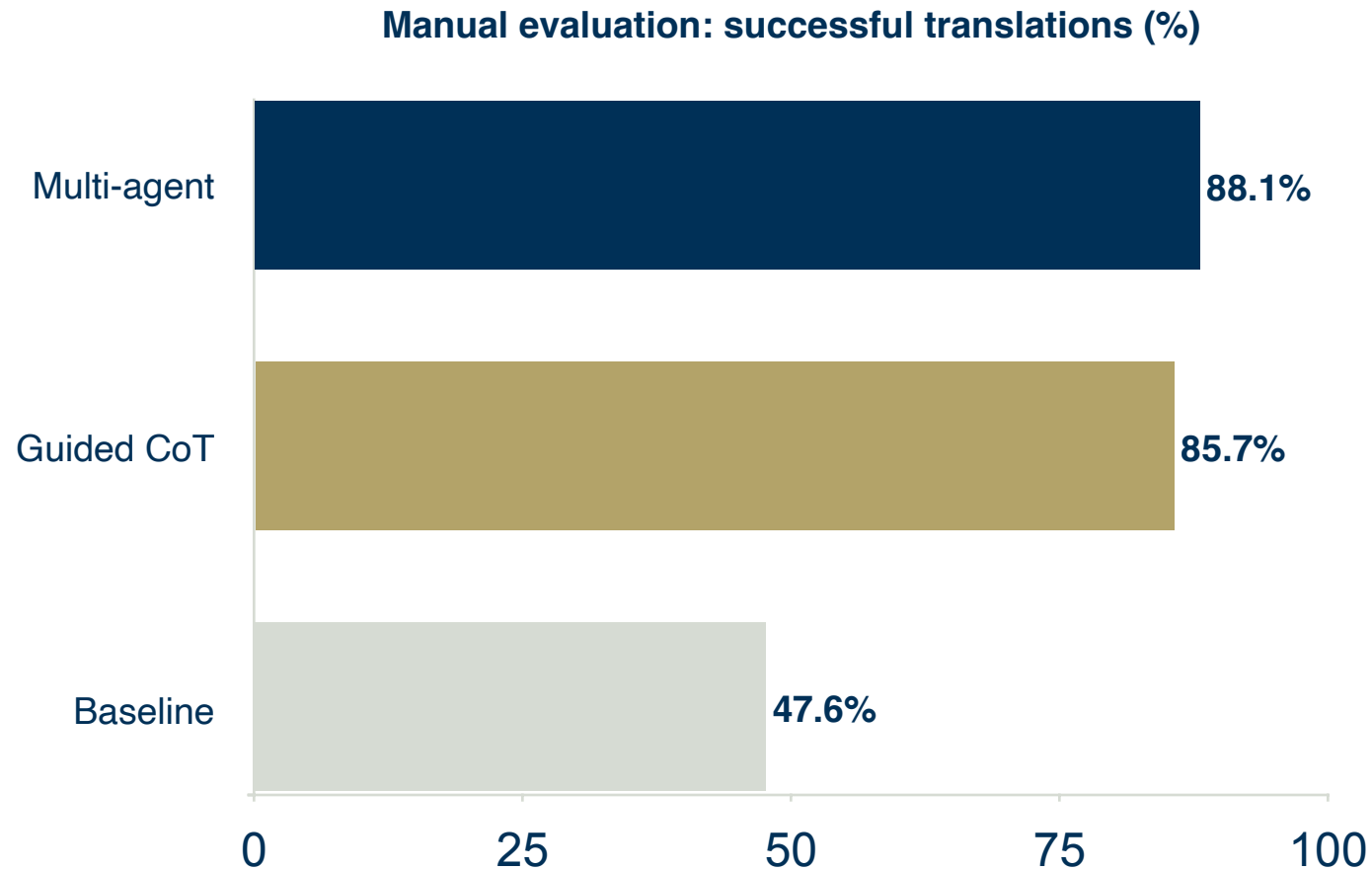
EVALUATORS (after MMTE)



Stop at mean score ≥ 2.0

or after 5 rounds; keep the best-scoring candidate

Results: 1st and 2nd of 51



37 / 42

multi-agent
successes

36 / 42

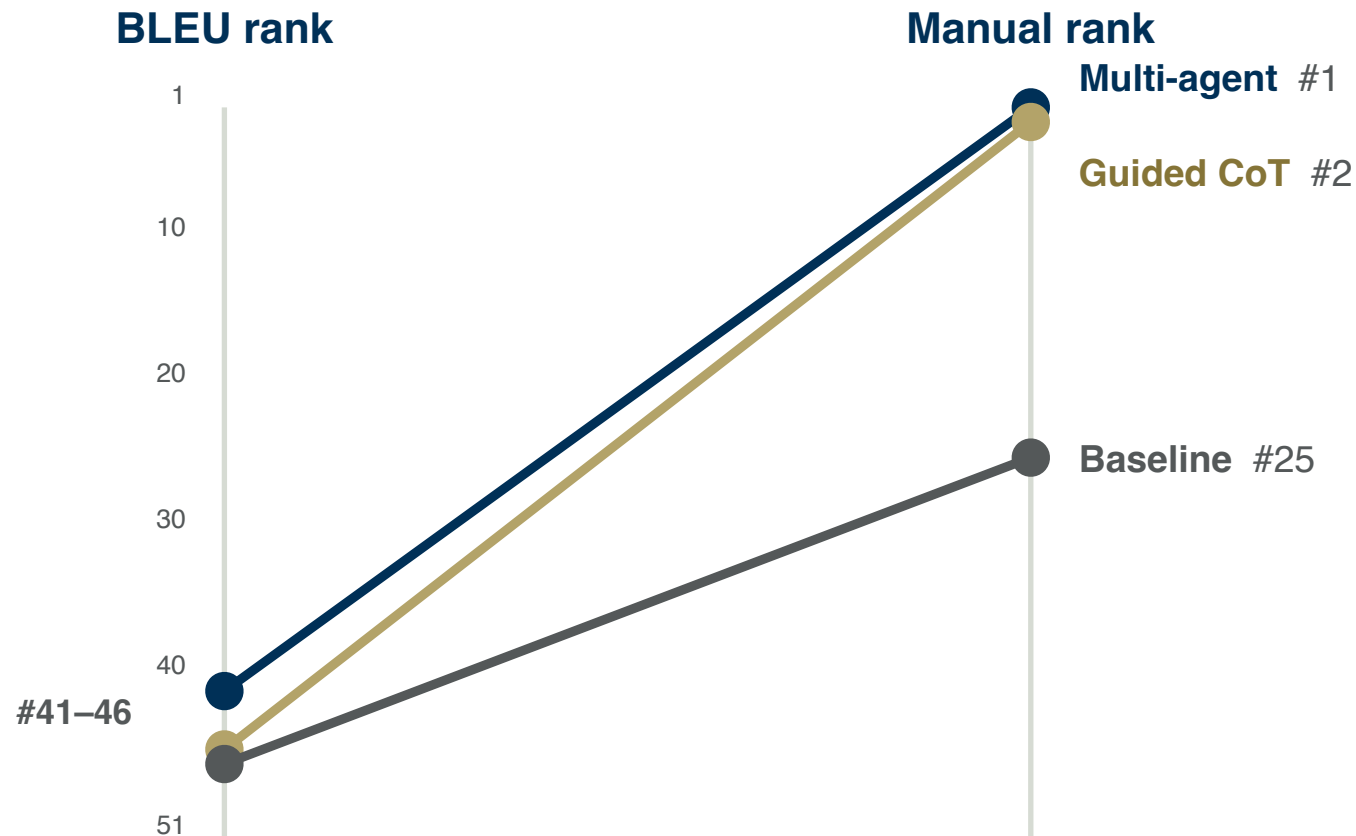
guided CoT
successes

Results: all metrics

System	BLEU	BERTScore	Pun loc.	Manual
Multi-agent	21.41 (41)	80.66 (41)	9.27% (1)	88.09% (1)
Guided CoT	16.52 (45)	78.42 (45)	7.85% (2)	85.71% (2)
Baseline	14.94 (46)	78.30 (46)	3.57% (45)	47.62% (25)

Rank among 51 submissions in parentheses.

The leaderboard flips: BLEU vs. human judges



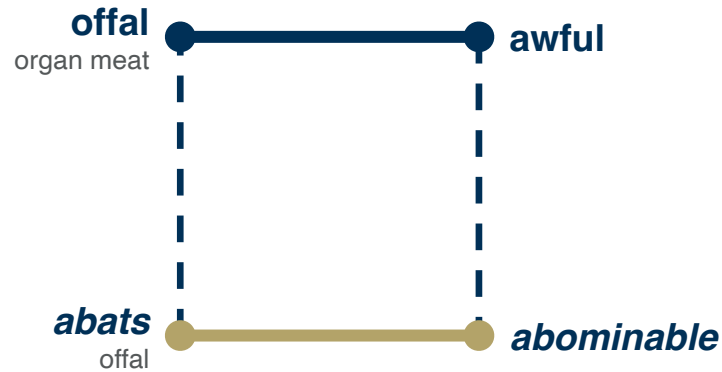
WHY

BLEU and BERTScore reward lexical overlap.

Good pun translation often has to give up that overlap to keep the joke.

Evaluate pun translation for functional equivalence, not lexical correspondence.

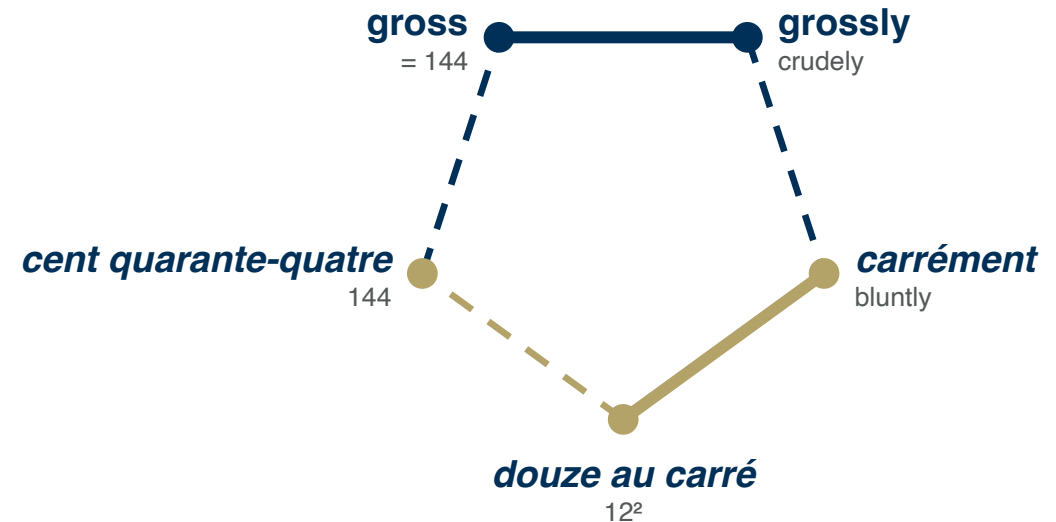
Our outputs read as Low's polygons



en_40 · multi-agent · square

« ... *des abats-ominables* »

abats + abominable: offal and awful in one word



en_160 · multi-agent · pentagon

« *Ça fait douze au carré* », dit-il carrément.

one leap: 144 → 12²; carré ≈ carrément closes the circle

— source pun - - translation - - semantic leap — target pun

What we learned

1

Functional equivalence beats lexical correspondence

Recreate the joke, not the words.

2

Automatic metrics reward the wrong objective

BLEU / BERTScore rank our systems 41st–46th.

3

Creativity before constraint

Generate freely, then guide and evaluate. Translation becomes search.

4

LLMs are surprisingly good evaluators

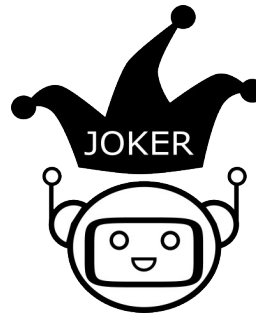
Simple run-time evaluation matched a heavily engineered pipeline.

5

Delabastita was right

Use the full available linguistic resources and textual leeway.

Thank you!



github.com/dsgt-arc/joker-2025 · {rdtaylorjr, bherbert6, msana3}@gatech.edu

Thanks to DS@GT ARC (Anthony Miyaguchi, Murilo Gustinelli), the CLEF JOKER 2025 organizers, and our reviewers.